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LAB REPORT

VAN de Graaff generator

➤ Background:

The concept of an electrostatic generator, where charge is mechanically transported in small amounts to the interior of a high-voltage electrode, could be traced back to the Kelvin water dropper which was invented by William Thomson (Lord Kelvin) in the year 1867.

In this device, charged water droplets fall into a bucket carrying the same charge polarity gradually increasing the charge. Here gravity moves the droplets against the opposing electrostatic field. Lord Kelvin was the first to suggest using a belt to carry charge instead of water.

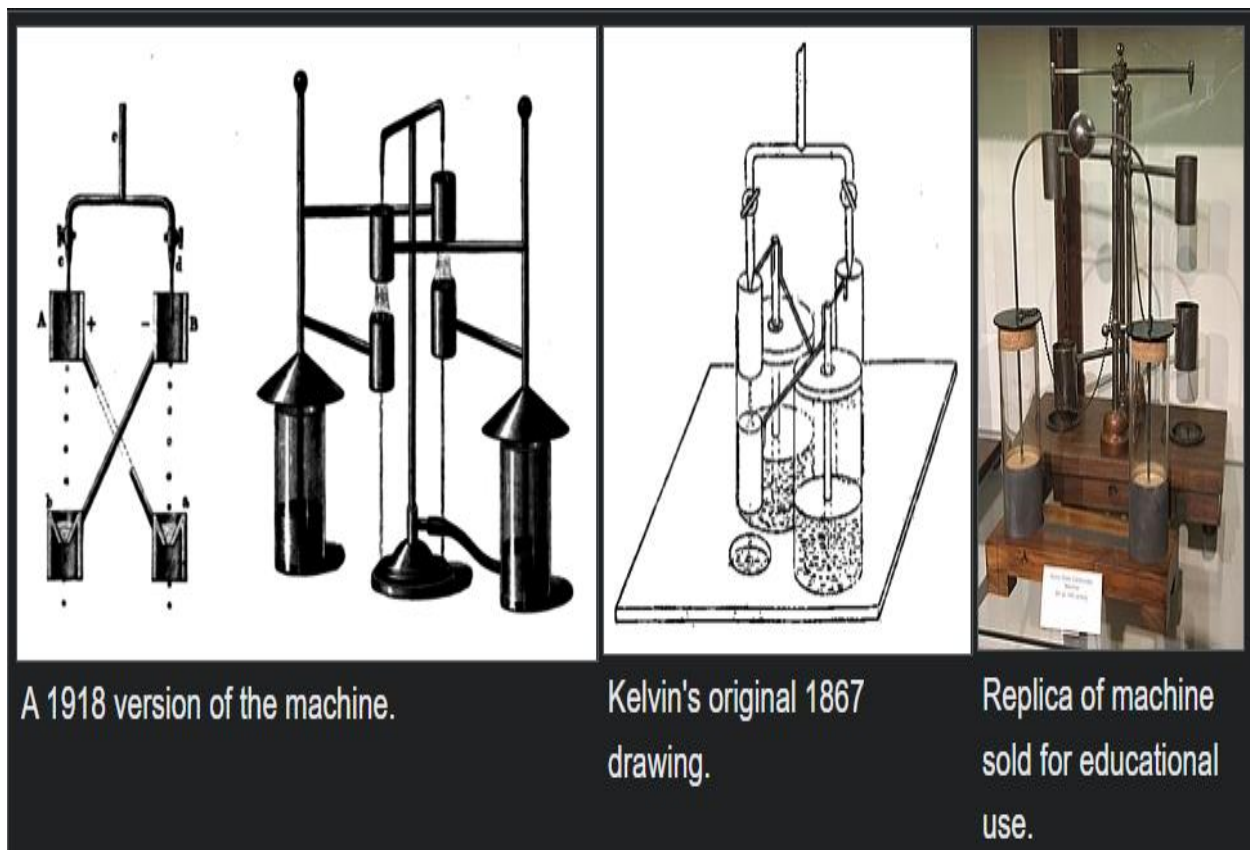


Figure: 1

Augusto Righi built the earliest electrostatic machine employing a continuous belt in the year 1872. Righi's design used a rubber belt with wire rings as charge carriers which

moved into a spherical metal electrode. The charge was transferred to the belt from the grounded lower roller via electrostatic induction from a charged plate.

Around the year 1890, John Gray also invented a similar machine. In the year 1903 Juan Burboa introduced a more complex version of it.

A significant influence on Van de Graaff's was a generator developed in the year 1920 by W.F. G. Swann, in which the charge was carried by falling metal balls.

➤ **Introduction:**

The Van de Graaff generator was invented by an American physicist Robert J. Van de Graaff in the year 1929. It is an electrostatic device that uses a moving belt to transport electric charge to a hollow metal sphere building up very high voltages.

Robert J. Van de Graaff with the help of his colleague Nicholas Burke, demonstrated the first model in October of the year 1929, achieving around 1.5 million volts—potentials that increased to 2.5 million volts when enclosed in insulating gases like sulfur hexafluoride (SF₆) to prevent electrical discharges.

➤ **Purpose:**

Originally it was developed as a particle accelerator for nuclear physics. It generates high voltages that could accelerate subatomic particles in an evacuated tube, providing critical insights into atomic energy and matter structure.

This technology was the most powerful of its kind until the invention of the cyclotron in the 1930s.

It continues to be used for generating particles and X-ray beams in nuclear research and medicine.

Open-air Van de Graaff generators can produce up to 5 megavolts, though modern enclosed versions in pressurized gas tanks achieve voltages as high as 25 megavolts. Small versions of the generator often produce around 100 kilovolts.

It is popular in educational and museum settings for demonstrations of electrostatic principles.

➤ **Principle:**

The Van de Graaff generator operates on the principle of “**electrostatic induction**” and two other phenomena which are given below.

- **Charge accumulation on a hollow conductor:** Any charge given to a hollow conductor is distributed uniformly on its outer surface.
- **Electrical discharge at sharp points:** Charged particles are readily discharged at pointed conductors.

➤ Construction:

The construction of a Van de Graaff generator involves several key components that work together to generate high-voltage static electricity. These components are as follows:

1. Base.
2. Motor.
3. Rubber belt.
4. 2-Rollers.
5. Spray comb.
6. Insulating rod.
7. Receiving comb.
8. Hollo sphere.
9. Power supply.
10. Discharge sphere (optional).

➤ Diagram:

Different diagrams of the Van De Graff Generator are given below:

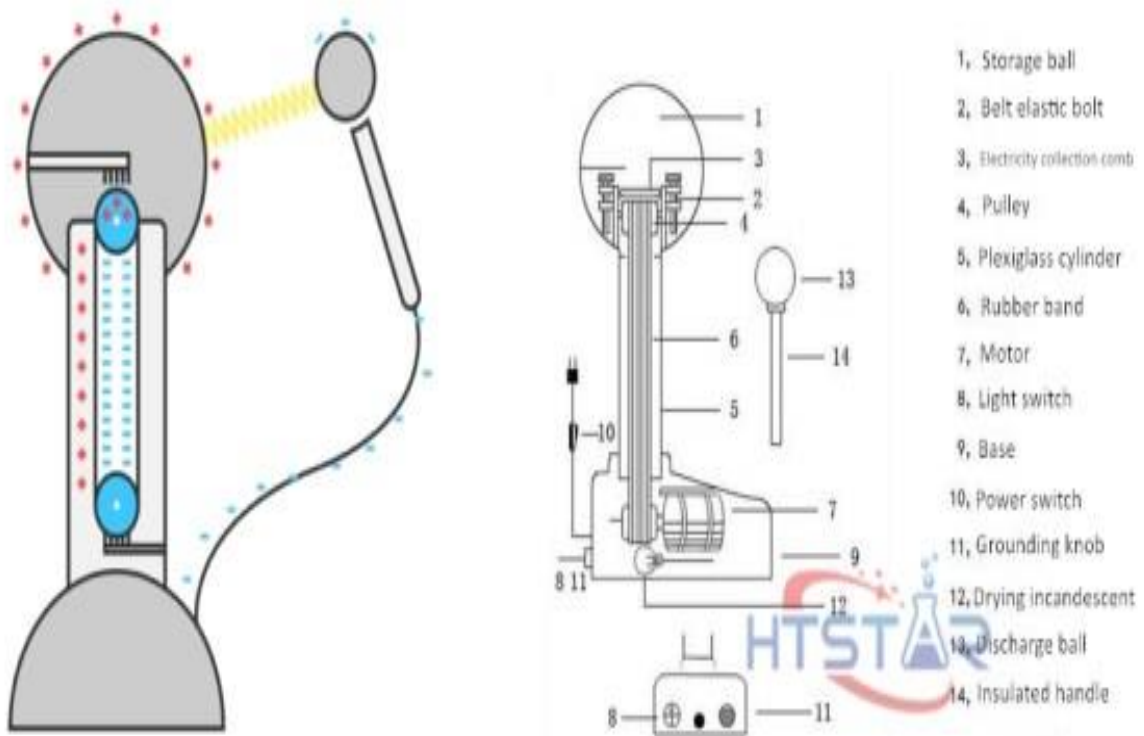


Figure: 1

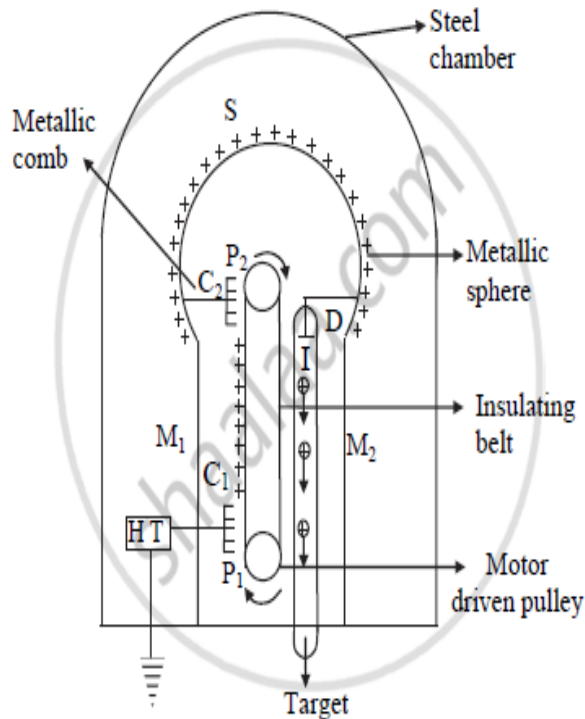


Figure: 2

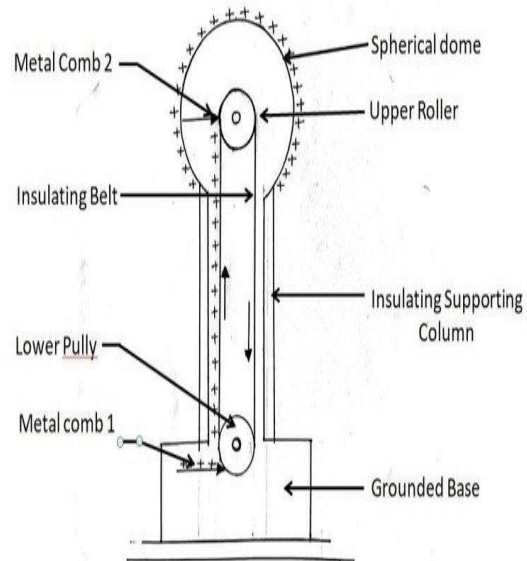


Figure: 3

➤ **Working:**

The Van de Graaff generator generates high voltage through a series of processes involving a moving belt:

1. **Charge Generation:** The rubber belt travels over two rollers made from different materials, acquiring electric charge from the spray comb through the triboelectric effect.
2. **Charge Transport:** The charged belt moves upward along an insulated rod, carrying the accumulated charge to the generator's receiving comb.
3. **Charge Transfer:** Metal combs with sharp points are positioned near the rollers. The lower comb deposits charge onto the belt, while the upper comb removes charge from the belt and transfers it to a large metal sphere.
4. **High Voltage Accumulation:** The charge builds up on the metal sphere, creating a very high voltage, often reaching several megavolts, resulting in strong electric fields.
5. **Discharge:** Once the voltage is sufficiently high, it can cause visible sparks or arcs to jump to nearby conductive objects.

This mechanism allows the Van de Graaff generator to produce high-voltage direct current (DC) electricity, making it valuable for educational demonstrations and research applications.

➤ **Applications:**

The Van de Graaff generator has a range of applications due to its ability to generate high-voltage electricity at low currents. Here are its primary uses:

1. **Educational Demonstrations:** Small Van de Graaff generators are widely used in classrooms and science museums to demonstrate principles of static electricity, including charge transfer, electric fields, and high-voltage discharges (sparks). They visually illustrate fundamental concepts in electrostatics.
2. **Particle Accelerators:** Originally designed as a particle accelerator, the Van de Graaff generator boosts particles to high energies. This application is still relevant in nuclear physics research, where generators are often housed in pressurized tanks with insulating gases to prevent electrical discharge and enhance performance.
3. **Nuclear Research:** In scientific research, Van de Graaff generators generate particle and X-ray beams. These high-energy beams are used to study nuclear reactions and atomic nuclei, contributing to advances in fields like particle physics and radiation research.
4. **Medical Applications:** In medical settings, Van de Graaff generators are valuable for nuclear medicine and radiation therapy. They generate powerful X-ray beams for imaging and treatment, supporting therapeutic applications in cancer treatment and other medical procedures.
5. **Industrial Testing and Calibration:** High-voltage testing is essential in electrical engineering. Van de Graaff generators are used to test the insulation of materials, calibrate equipment, and simulate high-voltage scenarios for industrial applications.